BMI Analysis from 2008-2012 in Scotland

1 Introduction

This project analyses the Body Mass Index (BMI) distributions in Scotland to investigate how BMI is influenced by a range of socio-economic and lifestyle factors. Using regression-based methods, the analysis aims to quantify the change in BMI over the period 2008 to 2012, as well as quantify the relationships between BMI and age, gender, socio-economic status and lifestyle factors. The overall goal is to assess changes in BMI with time and the effect of the factors on BMI, by testing the strength and significance of the resulting statistical model. Understanding these relationships is important for identifying key factors associated with BMI variation and informing potential public health insights.

1.1 Data Overview

The dataset contains observations collected between 2008 and 2012, including demographic, socio-economic and lifestyle variables such as age, education, sex, and fruit and vegetable consumtpion. The response variable in this analysis is BMI, which provides a quantitative measure for comparing individuals across time. The explanatory variables consist of the aforementioned lifestyle factors, along with age and sex, which are included to assess their potential influence on BMI.

2 Exploratory Analysis

We will first take a look at a summary statistic of the data in Table 1 and 2 below, to understand the data better.

Table 1: Summary for factor variables

Type	variable	n_missing	complete_rate	ordered	n_unique	counts
factor	Sex	0	1	FALSE	2	Fem: 13960, Mal: 11264
factor	Education	0	1	FALSE	6	Deg: 6891, No : 5298, Sta: 4845, Hig: 3717
factor	Veg	0	1	FALSE	2	Yes: 19797, No: 5427
factor	Fruit	0	1	FALSE	2	Yes: 17462, No: 7762

Table 2: Summary for numerical variables

Type	Variable	n_missing	comp_rate	mean	sd	p0	p25	p50	p75	p100	histogram
numeric	Age	0	1	50.0937	17.3254	16.0000	37.000	50.000	64.0000	103.0000	
numeric	BMI	0	1	27.8109	5.2573	12.5933	24.111	27.173	30.7875	59.1123	

From the tables we can see that the mean and median BMI is around 27 and the mean Age is 50.1 with a high standard deviation of 17.3. There are also more females than males in the data set and most individulas recorded consumption of recommended daily fruit and vegetable intake.

Next, to answer the first question of interest, we will look at a plot that shows changes in BMI over the years.

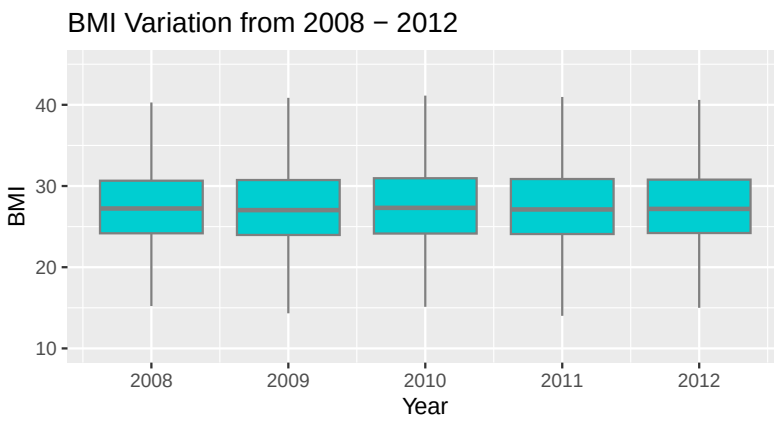

Figure 1: Change in BMI with Year

BMI distributions appear similar across years, suggesting that changes in BMI over time may not be prevalent.

To further explore relationships between lifestyle factors, a chi-squared test was conducted to assess the association between fruit and vegetable consumption. The results of the test indicated a strong evidence of association, suggesting that these variables capture related but distinct aspects of dietary behaviour. This supports their inclusion as predictors in subsequent regression modelling.

Table 3: Chi-square Test of Association between Fruit and Vegetable Intake

	Test	Chi.square	Df	P.value
X-squared	Chi-square test	1006.16	1	<0.001

This was built upon visually using a proportional bar chart (Figure 2), which illustrates the distribution of vegetable consumption across different levels of fruit intake. The plot suggests that there is a strong evidence of association between fruit and vegetable intake between these variables, with proportions varying across categories, indicating that individuals with different fruit consumption levels tend to have differing vegetable consumption patterns.

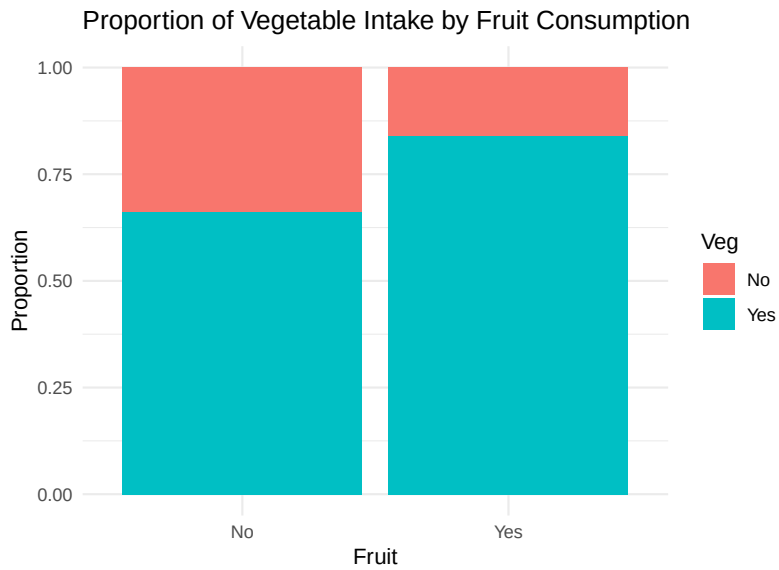

Figure 2: Bar Plot depicting the daily recommended Fruit & Vegetable intake

These findings justify the inclusion of both variables in the regression model, as they may capture related but distinct aspects of dietary behaviour.

Lastly, Figure 3 suggests a quadratic relationship between BMI and age: BMI increases with age up to middle age before gradually decreasing at older ages.

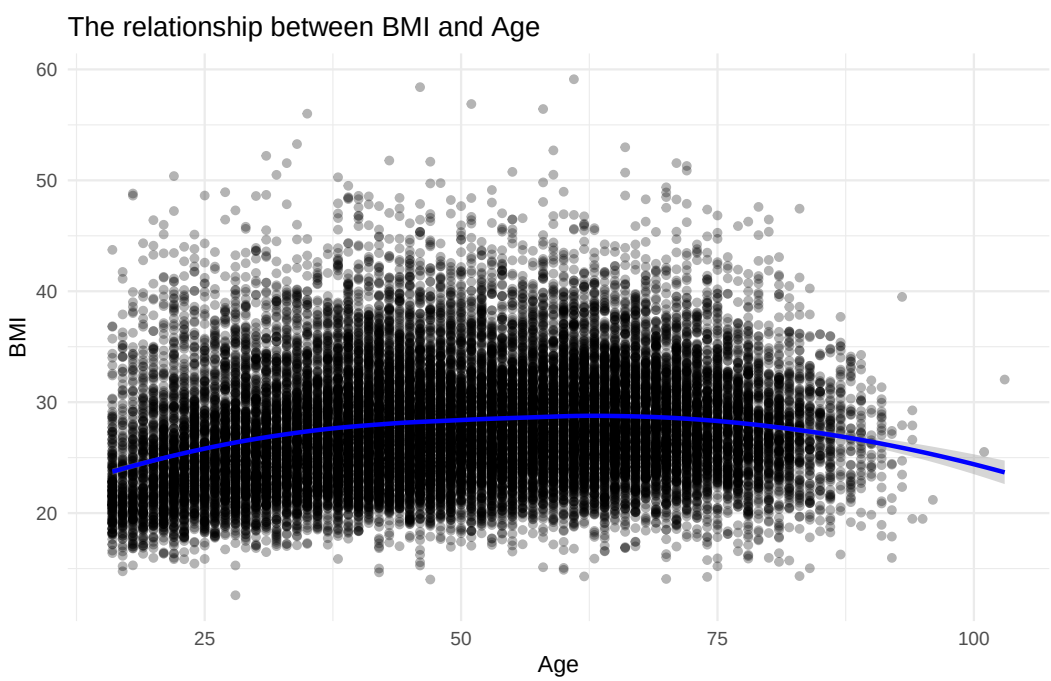

Figure 3: Scatterplot for BMI vs Age

Our initial analysis suggests that BMI does not change with year, but does change with age. There is also a correlation between fruit and vegetable intake behaviours. Next, we will conduct a formal analysis to further answer the two questions of interest.

3 Formal Analysis

3.1 Effect of Time on BMI

To investigate whether BMI changed over time, a linear model was fitted with year included as a continuous variable.

$$BMI_i = \beta_0 + \beta_1 Year_i + \varepsilon_i$$

Where β_1 represents the average change in BMI per year.

Table 4: Linear Trend of BMI over Time

Term	Estimate	F	P-value
Year	0.0236	0.91	0.34

The estimated coefficient for year is small and not statistically significant, suggesting that BMI did not change systematically over time during the study period. To further comment on the significance of Year, a 95% confidence interval was calculated to be [-0.03,0.07]; this contains zero, suggesting that Year is not a significant predictor in predicting BMI.

3.2 Multivariable Effects on BMI

Model Specification

To investigate how BMI varies with demographic and lifestyle factors, a multiple linear regression model was fitted with BMI as the response variable. The explanatory variables included age, sex, education level, fruit intake and vegetable intake.

Exploratory analysis suggested a non-linear relationship between BMI and age. To account for this, both age and its squared term were included in the model. In addition, selected interaction terms (age by sex, sex by education, and fruit by education) were included to analyse potential differences in effects across subgroups. Thus the final model is:

$$\begin{aligned} BMI_i = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Age}_i^2 + \beta_3 \cdot \mathbb{I}_{\{\text{Sex}\}}(\text{Male}) + \sum_{j=1}^{5} \beta_{4j} E_{ij} + \beta_5 \cdot \mathbb{I}_{\{\text{Fruit}\}}(\text{Yes}) + \beta_6 \cdot \mathbb{I}_{\{\text{Veg}\}}(\text{Yes}) \\ + \beta_7 (\text{Age}_i \times \text{Sex}_i) + \sum_{j=1}^{5} \gamma_j (\text{Sex}_i \times E_{ij}) + \sum_{j=1}^{5} \delta_j (\text{Fruit}_i \times E_{ij}) + \varepsilon_i \end{aligned}$$

where:

BMI_i : Body Mass Index for individual i

Age_i : Age (in years) for individual i

Sex_i : Indicator variable for sex of individual i

E_{ij} : Indicator variable for education level j of individual i

$Fruit_i$: Indicator variable for fruit intake of individual i

Veg_i : Indicator variable for vegetable intake of individual i

β_0 : Intercept term

β_1 : Coefficient representing the linear effect of age

β_2 : Coefficient representing the quadratic effect of age

β_3 : Coefficient representing the effect of sex

β_{4j} : Coefficients representing the effects of education levels j = 1,...,5

β_5 : Coefficient representing the effect of fruit intake

β_6 : Coefficient representing the effect of vegetable intake

β_7 : Coefficient representing the interaction between age and sex

γ_j : Coefficients representing the interaction between sex and education levels j

δ_j : Coefficients representing the interaction between fruit intake and education levels j

ε_i : Random error term for individual i

The Reference categories in the model are Sex = Female; Education = Baseline education category (reference level); Fruit = No fruit intake; Veg = No vegetable intake. All interaction terms are interpreted relative to these reference categories.

Model Selection

To assess whether including the interaction terms improve the model fit, a model without interactions was compared with the final model including interactions.

Table 5: Model Comparison: Effect of Including Interaction Terms

Comparison	Df	Sum.of.Squares	F.statistic	P.value
Without interactions vs With interactions	11	2164.2	7.54	<0.001

The comparison indicated that the inclusion of interaction terms improved the model fit, supporting their retention in the final model. Also on calculating the AIC, the model with interation terms was preferred as it minimised the AIC.

Results

The results of the multivariable model is shown in Table 6

Table 6: Estimate value and 95%CI

Term	Estimate	P-Value	Lower CI	Upper CI
Age	0.2976	0.0000	0.2772	0.3181
Age^2	-0.0026	0.0000	-0.0028	-0.0024
No qualifications	0.5941	0.0044	0.1850	1.0032
Fruit intake (Yes)	-0.4203	0.0060	-0.7198	-0.1208
Vegetable intake (Yes)	-0.3586	0.0000	-0.5168	-0.2004

From the confidence interval estimates, we can conclude that Age was positively associated with BMI, with an estimated increase of 0.30 BMI units per additional year, holding other variables constant. The quadratic age term was negative and statistically significant. This confirms a non-linear relationship between age and BMI, in which BMI increases with age at a decreasing rate. Individuals with no qualifications had an estimated 0.59 BMI units higher than the reference education group. Fruit intake was associated with a decrease of 0.42 BMI units compared to no fruit intake. Vegetable intake was associated with a decrease of 0.36 BMI units compared to no vegetable intake.

Also to check multicollinearity, we used the VIF function from library car. The result suggests that there is no multicollinearity between the variables since the adjusted VIF is less than 5, except for Age and Age^2. These can be ignored as they are the same variables.

Diagnostics Plots

Finally to assess the model fit, diagnostic plots from the function check_model were produced, as demonstrated in Figure 4.

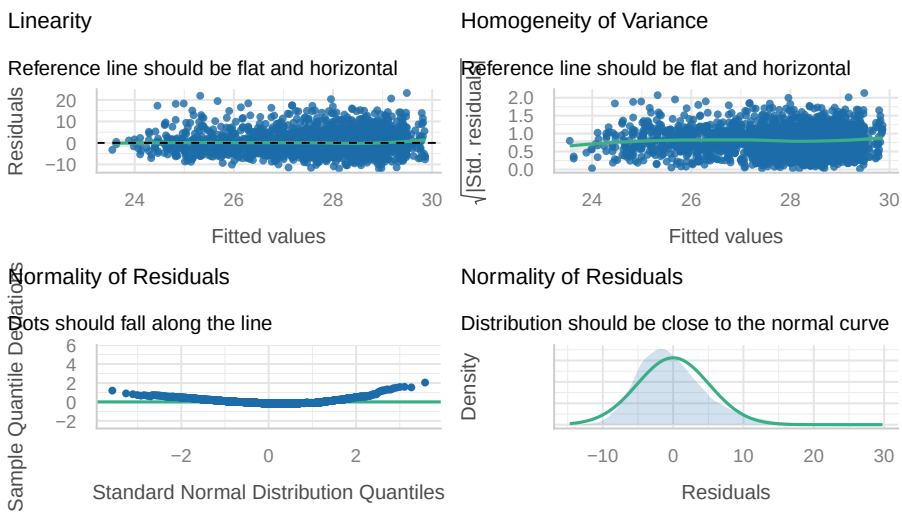

Figure 4: Model diagnostic plot

The diagnostic plots suggest that the assumptions of linearity and homoscedasticity are reasonably satisfied. Although the Q–Q plot shows some deviation from normality in the tails, this is not substantial given the large sample size. Overall, no major violations of model assumptions are evident.

4 Conclusion

In conclusion, this analysis investigated the relationship between BMI and a range of demographic and lifestyle factors using data from 2008 to 2012. The results indicate that there was no statistically significant change in BMI over time, suggesting that overall BMI levels remained relatively stable across the study period.

The analysis identified age as a key predictor of BMI, with evidence of a significant non-linear relationship. The inclusion of a quadratic age term improved model fit, indicating that the effect of age on BMI varies across different age groups rather than following a constant trend.

While dietary variables such as fruit and vegetable consumption showed evidence of association in the exploratory analysis and contributed to the modelling framework, variables such as sex were not found to be statistically significant once other factors were accounted for. This suggests that their explanatory power is limited within the context of this model.

Overall, the final model provides reasonable representation of the factors influencing BMI. Nevertheless, the presence of unexplained variability suggests that the model could be improved by incorporating more detailed measures of lifestyle and social-economic status, such as physical activity levels or income. Additionally, extending the model to include interaction terms or alternative non-linear structures may allow for a more accurate representation of underlying relationships.